```
SQL> CREATE TABLE EMPLOYEE_SALES (
  2      EMP_ID NUMBER,
  3      EMP_NAME VARCHAR2(50),
  4      DEPARTMENT VARCHAR2(30),
  5      SALARY NUMBER,
  6      SALES_MONTH VARCHAR2(20),
  7      SALES_AMOUNT NUMBER
  8  );

Table created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (101,'Amit','IT',60000,'January',50000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (102,'Priya','IT',75000,'February',65000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (103,'Rahul','IT',75000,'March',70000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (104,'Sneha','HR',55000,'January',30000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (105,'Kiran','HR',65000,'February',40000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (106,'Neha','HR',65000,'March',45000);
```

Command Prompt - sqlplus s

```
SQL> INSERT INTO EMPLOYEE_SALES VALUES (104,'Sneha','HR',55000,'January',30000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (105,'Kiran','HR',65000,'February',40000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (106,'Neha','HR',65000,'March',45000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (107,'Arjun','Sales',80000,'January',90000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (108,'Pooja','Sales',70000,'February',95000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (109,'Vikram','Sales',60000,'March',85000);

1 row created.

SQL> INSERT INTO EMPLOYEE_SALES VALUES (110,'Anjali','Finance',70000,'January',60000);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.

SQL>
```

```
SQL> SELECT EMP_ID,
  2         EMP_NAME,
  3         DEPARTMENT,
  4         SALARY,
  5         RANK() OVER (
  6             PARTITION BY DEPARTMENT
  7             ORDER BY SALARY DESC
  8         ) AS SALARY_RANK
  9  FROM EMPLOYEE_SALES;

    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                       SALARY SALARY_RANK
------------------------------ ---------- -----------
       110 Anjali
Finance                           70000           1

       105 Kiran
HR                                65000           1

       106 Neha
HR                                65000           1


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                       SALARY SALARY_RANK
------------------------------ ---------- -----------
       104 Sneha
HR                                55000           3

       103 Rahul
```

```
HR                                 65000           1

       106 Neha
HR                                 65000           1


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY SALARY_RANK
------------------------------ ---------- -----------
       104 Sneha
HR                                 55000           3

       103 Rahul
IT                                 75000           1

       102 Priya
IT                                 75000           1


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY SALARY_RANK
------------------------------ ---------- -----------
       101 Amit
IT                                 60000           3

       107 Arjun
Sales                              80000           1

       108 Pooja
Sales                              70000           2


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY SALARY_RANK
------------------------------ ---------- -----------
       109 Vikram
Sales                              60000           3

10 rows selected.
```

```
SQL> SELECT EMP_ID,
  2         EMP_NAME,
  3         DEPARTMENT,
  4         SALARY
  5  FROM (
  6      SELECT EMP_ID,
  7             EMP_NAME,
  8             DEPARTMENT,
  9             SALARY,
 10             DENSE_RANK() OVER (
 11                 PARTITION BY DEPARTMENT
 12                 ORDER BY SALARY DESC
 13             ) AS SALARY_RANK
 14      FROM EMPLOYEE_SALES
 15  )
 16  WHERE SALARY_RANK = 1;

    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY
------------------------------ ----------
       110 Anjali
Finance                         70000

       105 Kiran
HR                              65000

       106 Neha
HR                              65000


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY
------------------------------ ----------
       102 Priya
IT                              75000

       103 Rahul
IT                              75000
```

```
    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY
------------------------------ ----------
       110 Anjali
Finance                         70000

       105 Kiran
HR                              65000

       106 Neha
HR                              65000


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY
------------------------------ ----------
       102 Priya
IT                              75000

       103 Rahul
IT                              75000

       107 Arjun
Sales                           80000


6 rows selected.

SQL>
```

Command Prompt - sqlplus s

```
SQL> SELECT EMP_NAME,
  2         SALES_MONTH,
  3         SALES_AMOUNT,
  4         LAG(SALES_AMOUNT) OVER (
  5             PARTITION BY EMP_NAME
  6             ORDER BY SALES_MONTH
  7         ) AS PREVIOUS_SALES
  8  FROM EMPLOYEE_SALES;

EMP_NAME                                           SALES_MONTH
-------------------------------------------------- --------------------
SALES_AMOUNT PREVIOUS_SALES
------------ --------------
Amit                                               January
       50000

Anjali                                             January
       60000

Arjun                                              January
       90000


EMP_NAME                                           SALES_MONTH
-------------------------------------------------- --------------------
SALES_AMOUNT PREVIOUS_SALES
------------ --------------
Kiran                                              February
       40000

Neha                                               March
       45000

Pooja                                              February
       95000


EMP_NAME                                           SALES_MONTH
-------------------------------------------------- --------------------
SALES_AMOUNT PREVIOUS_SALES
------------ --------------
Priya                                              February
       65000

Rahul                                              March
       70000

Sneha                                              January
       30000


EMP_NAME                                           SALES_MONTH
-------------------------------------------------- --------------------
SALES_AMOUNT PREVIOUS_SALES
------------ --------------
Vikram                                             March
       85000


10 rows selected.
```

10 rows selected.

```
SQL> SELECT EMP_ID,
  2         EMP_NAME,
  3         DEPARTMENT,
  4         SALARY,
  5         NTILE(4) OVER (
  6             ORDER BY SALARY DESC
  7         ) AS SALARY_GROUP
  8  FROM EMPLOYEE_SALES;
```

EMP_ID	EMP_NAME	DEPARTMENT	SALARY	SALARY_GROUP
107	Arjun	Sales	80000	1
102	Priya	IT	75000	1
103	Rahul	IT	75000	1

EMP_ID	EMP_NAME	DEPARTMENT	SALARY	SALARY_GROUP
108	Pooja	Sales	70000	2
110	Anjali	Finance	70000	2
106	Neha	HR	65000	2

EMP_ID	EMP_NAME	DEPARTMENT	SALARY	SALARY_GROUP
105	Kiran	HR	65000	3
101	Amit	IT	60000	3
109	Vikram	Sales	60000	4

EMP_ID	EMP_NAME	DEPARTMENT	SALARY	SALARY_GROUP
104	Sneha	HR	55000	4

```
SQL> SELECT EMP_ID,
  2         EMP_NAME,
  3         DEPARTMENT,
  4         SALARY,
  5         AVG(SALARY) OVER (
  6             PARTITION BY DEPARTMENT
  7         ) AS DEPT_AVG_SALARY,
  8         SUM(SALARY) OVER (
  9             PARTITION BY DEPARTMENT
 10         ) AS DEPT_TOTAL_SALARY
 11  FROM EMPLOYEE_SALES;

    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY DEPT_AVG_SALARY DEPT_TOTAL_SALARY
------------------------------ ---------- -------------- -----------------
       110 Anjali
Finance                         70000          70000             70000

       105 Kiran
HR                              65000      61666.6667            185000

       106 Neha
HR                              65000      61666.6667            185000


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY DEPT_AVG_SALARY DEPT_TOTAL_SALARY
------------------------------ ---------- -------------- -----------------
       104 Sneha
HR                              55000      61666.6667            185000

       101 Amit
IT                              60000          70000            210000

       103 Rahul
IT                              75000          70000            210000


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY DEPT_AVG_SALARY DEPT_TOTAL_SALARY
```

```
       106 Neha
HR                                65000      61666.6667            185000


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY DEPT_AVG_SALARY DEPT_TOTAL_SALARY
------------------------------ ---------- -------------- -----------------
       104 Sneha
HR                                55000      61666.6667            185000

       101 Amit
IT                                60000           70000            210000

       103 Rahul
IT                                75000           70000            210000


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY DEPT_AVG_SALARY DEPT_TOTAL_SALARY
------------------------------ ---------- -------------- -----------------
       102 Priya
IT                                75000           70000            210000

       109 Vikram
Sales                             60000           70000            210000

       107 Arjun
Sales                             80000           70000            210000


    EMP_ID EMP_NAME
---------- --------------------------------------------------
DEPARTMENT                     SALARY DEPT_AVG_SALARY DEPT_TOTAL_SALARY
------------------------------ ---------- -------------- -----------------
       108 Pooja
Sales                             70000           70000            210000


10 rows selected.

SQL>
```